DIGITAL FORENSICS SESSION

Digital Forensics:
From Digital Evidence to
Cyber Investigation

A technical walkthrough of how investigators identify, preserve, and analyze digital evidence to reconstruct real cyber incidents.

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CASE FILE

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STATUS

ACTIVE

SCOPE

IITDU

WHO AM I??

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- ➢ Certified Network Security Practitioner (CNSP)
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What is Forensics?

The scientific process of examining evidence to establish facts admissible in a legal or professional context.

Purpose of Investigation

Apply structured, scientific methods to answer: what happened, when, how, and who was responsible — backed by verifiable evidence.

Core Principle

Findings must be objective, repeatable, and defensible — every conclusion traces back to preserved, untampered evidence.

Evidence
01

Raw material collected from the scene

Analysis
02

Systematic examination using proven methods

Findings
03

Objective observations drawn from analysis

Conclusion
04

Defensible answer to what actually happened

What is Digital Forensics?

The application of forensic science to digital devices and data — uncovering, preserving, and interpreting electronic evidence.

Digital Evidence

Any information of probative value stored or transmitted in binary form — files, logs, memory, network traffic, and more.

It must be handled so that it stays legally and technically trustworthy — this is why digital forensics follows a strict, repeatable process.

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CORE STAGES	CHAIN OF CUSTODY	TOLERANCE FOR TAMPERING

- **Identification**
- **Preservation**
- **Acquisition**
- **Examination**
- **Analysis**
- **Reporting**
- **Documentation — recorded across every stage above**

Forensics vs. Digital Forensics

Same investigative principles — applied to a fundamentally different kind of evidence.

Traditional Forensics

- **Physical evidence**
  Fingerprints, fibers, blood, weapons
- **Fixed in space**
  Tied to a physical crime scene
- **Visual examination**
  Microscopes, chemical tests, ballistics
- **Degrades naturally**
  Evidence can decay or be destroyed

VS

Digital Forensics

- **Digital evidence**
  Files, logs, memory, network packets
- **Can exist anywhere**
  Local disks, cloud, mobile, remote servers
- **Technical examination**
  Hashing, imaging, specialized tooling
- **Fragile & volatile**
  Can be altered or lost by a single reboot

Types of Digital Forensics

Ten specialized domains, each focused on a different source of digital evidence.

Computer / Disk File systems & storage	**Memory** Volatile RAM artifacts	**Network** Traffic & packet capture	**Mobile** Phones & tablets	**Cloud** SaaS & remote storage
Database Structured record stores	**Email** Headers & message content	**Browser / Web** History & web artifacts	**Malware** Malicious code analysis	**IoT / Embedded** Smart & embedded devices

Digital Forensics Investigation Model

A seven-stage chain of custody that keeps evidence sound from first contact to final report.

Identification

Spot potential evidence sources

Preservation

Protect evidence from change

Collection

Acquire forensic copies

Examination

Extract relevant data

Analysis

Interpret what the data shows

Documentation

Record every action taken

Reporting

Present findings clearly

Chain of custody must remain unbroken across all seven stages — any gap can make evidence inadmissible.

Digital Evidence

What actually counts as evidence — anything that can help reconstruct events on a device or network.

Files
Documents, images, archives

Metadata
Hidden file properties

Logs
System & application events

Browser History
Sites visited & downloads

Emails
Headers, content, attachments

Network Packets
Captured traffic on the wire

Memory Artifacts
Running processes & keys in RAM

Deleted Files
Recoverable remnants on disk

Mobile Artifacts
App data, messages, GPS

Cloud Data
Synced files & account activity

Common Digital Forensics Formats

The container formats investigators encounter when imaging disks, capturing traffic, or dumping memory.

RAW / DD

Bit-for-bit disk image with no metadata — the universal baseline format.

E01 (EWF)

Compressed image with built-in hashing, used by EnCase & FTK.

AFF / AFF4

Open, extensible forensic container with embedded metadata.

VHD / VHDX

Microsoft virtual disk formats, common in Hyper-V environments.

VMDK

VMware virtual disk format encountered in server & cloud hosts.

QCOW2

QEMU/KVM virtual disk format used in Linux virtualization.

PCAP / PCAPNG

Captured network traffic for protocol & packet analysis.

Memory Dumps

Raw RAM capture (.mem/.raw) for volatile memory analysis.

ZIP / Archives

Compressed evidence bundles for storage & transfer.

Common Digital Forensics Tools

The core toolkit investigators reach for across disk, memory, network, and mobile analysis.

Autopsy Disk & filesystem GUI	**The Sleuth Kit** Disk forensics library	**FTK Imager** Evidence acquisition	**Magnet AXIOM** Multi-source artifacts	**EnCase** Enterprise investigation	**X-Ways Forensics** Disk & file analysis	**Volatility** Memory forensics
Wireshark Network / PCAP analysis	**KAPE** Rapid artifact collection	**Plaso / log2timeline** Timeline creation	**Eric Zimmerman Tools** Windows artifact parsing	**Cellebrite** Mobile device forensics	**Ghidra** Reverse engineering	**YARA** Malware & pattern ID

Most Common Software & Their Uses

A quick-reference matrix mapping each tool to its primary investigative role.

SOFTWARE	PRIMARY USE	SOFTWARE	PRIMARY USE
Autopsy	Disk & filesystem investigation	**Wireshark**	Network / PCAP analysis
FTK Imager	Evidence acquisition & imaging	**KAPE**	Rapid artifact collection
Magnet AXIOM	Computer / mobile / cloud artifacts	**Plaso**	Timeline creation
EnCase	Enterprise forensic investigation	**Cellebrite**	Mobile device forensics
X-Ways	Disk and filesystem forensics	**Ghidra**	Reverse engineering
Volatility	Memory forensics	**YARA**	Malware / artifact identification

Mini Investigation: Employee Laptop Compromise

A realistic, fictional incident walked through the full forensic pipeline — start to finish.

01
Incident
Unusual outbound traffic flagged

02
Evidence Collection
Laptop isolated & secured

03
Disk Image
Forensic copy acquired

04
Timeline
Activity reconstructed

05
USB / Browser Artifacts
Data exfil traces found

06
Malware Discovery
Malicious binary identified

07
Analysis
Behavior & scope confirmed

08
Report
Findings delivered to IR team

Where Digital Forensics is Used

From courtrooms to corporate SOCs — digital forensics underpins modern security and justice.

Cybercrime Investigation	Incident Response	Corporate Investigations	Financial Fraud	Insider Threats
Malware Investigations	Data Breach Investigations	Law Enforcement	E-Discovery	National Security

Real-Life Digital Forensics Jobs

Career roles across the DFIR ecosystem — from front-line SOC work to specialized examination.

- Digital Forensic Analyst
- DFIR Analyst
- Incident Response Analyst
- Malware Analyst
- Memory Forensics Analyst
- Mobile Forensic Examiner
- Cybercrime Investigator
- SOC Analyst → DFIR Pathway
- Threat Hunter
- Forensic Consultant
- Digital Forensics Expert

Career Roadmap

A step-by-step path from foundational IT skills to a career in DFIR.

01 **Networking + Linux + Windows**

02 **Cybersecurity Fundamentals**

03 **Digital Forensics Fundamentals**

04 **Disk / Memory / Network / Mobile Forensics**

05 **Hands-on Labs & CTFs**

06 **Certifications / Portfolio**

07 **DFIR / Digital Forensics Career**

Skills You Need

The technical and analytical foundation every digital forensics investigator builds on.

- Windows Internals
- Linux
- Networking
- Filesystems
- OS Artifacts
- Python
- PowerShell / Bash
- Malware Fundamentals
- Timeline Analysis
- Log Analysis
- Report Writing
- Critical Thinking

How to Start Learning

A practical path from zero to your first real forensic investigation.

01

Build a forensic lab

Set up a safe, isolated practice environment

02

Analyze disk images

Practice with publicly available sample images

03

Practice memory forensics

Work through RAM dumps with Volatility

04

Analyze PCAP files

Study captured traffic in Wireshark

05

Investigate Windows artifacts

Explore registry, event logs, prefetch

06

Solve forensic CTF challenges

Apply skills under realistic pressure

07

Learn the core toolkit

Autopsy, FTK Imager, Volatility, Wireshark

08

Document your investigations

Build a portfolio of write-ups

KEY TAKEAWAY

“Digital Forensics is not just about finding files — it is about reconstructing what happened from digital evidence.”

Preserve Evidence

Protect integrity from first contact to final report

Analyze Systematically

Follow a repeatable, defensible process every time

Tell the Story with Evidence

Let the data — not assumptions — drive the conclusion

Questions?

Zaber Mahmud | EWU Cyber Security Club

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References & Further Reading

Authoritative sources for continued study in digital forensics and DFIR.

- NIST — Digital Forensics publications (Computer Forensics Tool Testing Program)
- NIST Computer Security Resource Center (csrc.nist.gov)
- SWGDE — Scientific Working Group on Digital Evidence
- The Sleuth Kit & Autopsy official documentation
- Volatility Foundation — memory forensics documentation
- Wireshark official documentation & user guide
- Magnet Forensics — knowledge base & resources
- Cellebrite — mobile forensics resources
- CISA — Cybersecurity & Infrastructure Security Agency resources